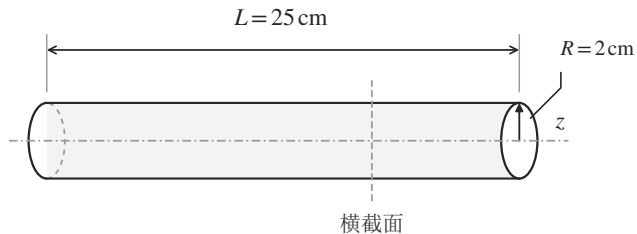


(a) 圆柱几何与径向简化



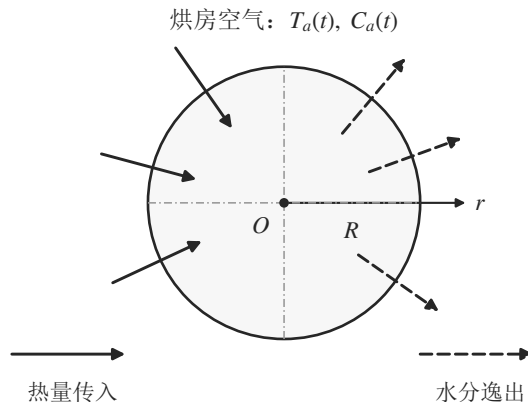
$$T = T(r, t), \quad C = C(r, t)$$

轴对称；忽略轴向梯度和端面传递

初始状态

$$T_0 = 301.15 \text{ K } (28^\circ\text{C}), \quad C_0 = 2.55 \text{ kg/kg}$$

(b) 横截面与传递方向



$$\text{轴线对称: } \partial_r T|_0 = \partial_r C|_0 = 0$$

侧壁边界（径向向外为正）

$$-k \frac{\partial T}{\partial r} \Big|_R = h[T(R, t) - T_a(t)]$$

$$-D \frac{\partial C}{\partial r} \Big|_R = h_m[C(R, t) - C_a(t)]$$

箭头示升温失水工况；横截面放大绘制，传质边界采用题设经验口径。